MILITARY SPECIFICATION

TRANSCEIVER, VHF/UHF

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a simplex portable/mobile transceiver designed to operate within the following frequency ranges and associated modulations:

a.	116 to 137 megahertz (MHz)	Amplitude modulation
b.	138 to 150.8 MHz	Frequency modulation
c.	225 to 400 MHz	Amplitude modulation

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore, General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories And Repair Parts, Packaging And Packing Of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definition Of And Basic Requirement For Enclosure For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5820

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MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The equipment shall be in accordance with the provisions of MIL-E-16400 to the extent specified herein.

3.1.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

3.2 Physical characteristics.

3.2.1 Configuration. The units comprising a complete system shall be designed for installation and operation as a vehicular transceiver, for operation as a self-sufficient portable transceiver, and for installation and operation as a fixed base station transceiver.

3.2.2 Modular construction. The equipment shall employ modular type construction to the extent that the circuitry for each functional division of the circuit shall be incorporated in a separate removable electronic assembly or subassembly.

3.2.3 Weight. The weight of the complete equipment shall not exceed 13.61 kilograms (kg) (30 pounds (lbs)).

3.2.4 Size. The dimensions of the complete equipment shall not exceed 15.24 centimeters (cm) (6 inches(in)) in height, 35.56 cm (14 in) in width, and 30.48 cm (12 in) in depth.

3.2.5 Front panel. The front panel shall contain all controls and instrumentation necessary for operation of the transceiver. As a minimum, these shall be: frequency selector, preset frequency selector, off-on-charge switch, volume control, squelch control, power ON light, transmitter ON light, fuse access, handset connector, and loudspeaker disable switch.

3.2.5.1 Controls, indicators, and panel layouts. The design of controls and indicators, and their arrangement on operator and maintainer panels, shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the controls, indicators and panel layout requirements of MIL-E-16400.

3.3 Parts, materials, and processes.

3.3.1 Material. The material and components used in the construction of the equipment shall be the highest quality consistent with the proposed use and performance specified herein. Metal parts shall be inherently resistant to the corrosive action of a moist sea atmosphere, or shall be protected by means of chemical or electrolytic finishes and treatments, organic or electro-deposited metallic coatings, or a combination thereof.

3.3.2 Dissimilar metals. Dissimilar metals in contact shall be avoided. Unavoidable contact between dissimilar metals shall be adequately protected against galvanic corrosion.

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3.3.3 Components.

3.3.3.1 Active devices. All active circuits (amplifiers, oscillators, and so forth) shall employ solid state devices (diodes, transistors, and so forth). Use of thermionic electron tubes is not permissible.

3.3.3.2 Selection. The performance of the equipment shall not be dependent on the selection of individual resistors, capacitors, transistors, or other parts. Selection and prescreening of electronic and electrical parts and materials shall be in accordance with MIL-STD-965, Procedure I.

3.3.3.3 Interchangeability. For like equipments, all essential parts, including removable units, repair parts, and other items shall be interchangeable physically and electrically with corresponding items or parts without alteration or use of undue force. The equipment design shall include tolerances sufficient to accomodate various articles such as transistors, resistors, capacitors, and other parts involved having limited dimensions and without departure from the specified performance.

3.3.3.4 Arrangement. The arrangement of parts shall be such that replacement or adjustment of any part is possible without removal of or damage to adjacent parts. All parts shall be readily accessible for replacement or repair.

3.3.3.5 Protection. The transmitter final amplifier shall not be damaged under variable load conditions such as grounding, partial grounding, or disconnecting of the antenna. The receiver shall not be damaged by strong voltage (10 volts root-mean-square (rms)) applied at the antenna connector.

3.3.4 Cabinet.

3.3.4.1 Mechanical construction. The complete transceiver shall be contained in a drip-proof metal cabinet and shall be capable of passing the test for such an enclosure as specified in MIL-STD-108 (see 4.5.10). The equipment shall conform to the salt-fog requirement of MIL-E-16400 for sheltered equipment. A cover shall be provided with stowage for handset, power cables, and detachable antennas. The cover shall also protect the controls and other components on the front panel.

3.3.4.2 Identification (ID) plates and marking. Identification plates and marking shall conform to MIL-E-16400.

3.3.4.3 Finish. All metal surfaces of chassis and enclosures shall be protected against corrosion by chemical treatment, plating, or painting. The equipment cabinet shall be finished and painted light gray enamel or an approved equivalent, conforming to the requirements of MIL-E-16400.

3.3.5 Mounting and installation.

3.3.5.1 Vehicular mounting. A vehicular mounting bracket shall be furnished of such design as to provide quick removal and reinstallation of unit to and from an above flat-mounted surface or a below-mounted (dashboard) position with the use of tools after bracket installation.

3.3.5.2 Changeover. The equipment shall be designed so that it can be converted in operation from either a portable to a vehicular radio set or from a vehicular to a portable in five minutes on any primary power within its capabilities. The time required for original installation of mounting bracket, vehicular antenna, power and other cabling is not part of this requirement.

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3.3.5.3 Portable configuration. When operated as a portable equipment, the transceiver shall be mountable on a backpack or carried via a strap over the shoulder.

3.3.6 Power cables. The alternating current (AC) power cables shall be 182.88 cm to 274.32 cm (6 to 9 feet (ft)) in length, which shall be three conductor - grounding type with standard three-prong plug. The direct current (DC) cable(s) shall be 365.76 cm (12 ft) long stranded, insulated and weatherproof, coded red for positive and black for negative.

3.3.7 Antennas. The equipment shall be furnished with two sets of antennas. One set for portable or temporary base operation and one set for vehicular operation.

3.3.7.1 Portable antennas. The portable temporary base antenna shall operate as a vertical antenna connected to the antenna connector on the front panel. A 90 degree elbow coupler or similar device shall be furnished to enable vertical antenna operation with the equipment either on its back or bottom.

3.3.7.2 Vehicular antennas. The vehicular antennas shall be furnished with a temporary mount, gutter-clip, magnetic, or other. The coaxial transmission line shall be 304.80 cm to 365.76 cm (10 to 12 ft) in length with mating connector to the equipment antenna connector.

3.3.8 Soldering. Soldering shall be in accordance with Requirement 5 of MIL-STD-454, as specified in MIL-E-16400.

3.3.9 Handset. The handset shall be of the type H-189/U. Hang-up brackets shall be provided for ease of microphone access or replacement from any operating position.

3.3.10 Fusing. The transceiver shall be protected against overload by fuses or circuit-breakers.

3.4 Environmental conditions. The transceiver shall operate in accordance with the performance requirements specified herein under the conditions specified in 3.4.1 through 3.4.4.

3.4.1 Operating temperature. The operating temperature shall be any ambient temperature between the limits of -30° Celsius (C) to +60°C.

3.4.2 Humidity. The equipment shall be capable of operating in relative humidities ranging up to 95 percent for both continuous and intermittent periods.

3.4.3 Vibration. No fixed part shall become loose, nor any movable part shifted in position or adjustment, during the vibration cycles specified in 4.5.7. While vibrating the equipment shall conform to all electrical requirements at an ambient temperature between +20°C and +30°C and relative humidity between zero and 90 percent.

3.4.4 Mechanical shock. The equipment shall conform to all electrical requirements and shall suffer no mechanical damage after being subjected to a series of not less than 10 impacts in each plane (total 30). Each impact shall be not less than 20g acceleration (see 4.5.8).

3.5 Electrical and performance characteristics.

3.5.1 Primary power supply. The equipment shall be designed to operate from a primary single-phase AC line voltage source of 105 volts to 130 volts at a frequency of 57 to 63 hertz (Hz). The transceiver shall also be capable of operation from a DC line voltage source of 11.0 volts to 15.0 volts and from a DC line voltage source of 23.8 volts to 29.0 volts. All of the electrical and performance requirements specified herein shall be conformed to while operating from each of the sources specified herein.

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3.5.2 Receiver.

3.5.2.1 Receiver frequency range. The receiver shall be designed for amplitude modulation (AM) and frequency modulation (FM) operation at those frequencies listed in 1.1. Twenty-five kilohertz (kHz) channel selection increments are to be used in all ranges. All receiver specification requirements shall be conformed to for every frequency in every range.

3.5.2.2 Noise reduction. The receiver shall have a noise silencer or noise limiting circuit which eliminates ignition and other electrical noise from nearby vehicles.

3.5.2.3 RF input. The radio frequency (RF) input circuit of the receiver shall be designed to match a nominal 50-ohm unbalanced transmission line.

3.5.2.4 Receiver output. The receiver shall have two outputs. One output shall be connected to a built-in permanent magnet loudspeaker and the other shall be connected to the handset (see 3.3.9), the loudspeaker being disabled by a front-panel switch.

3.5.2.5 Handset audio output level. The audio power output level at the handset connector when terminated in a 500-ohm resistive load, shall be 20 to 25 decibels (dB) below that of the loudspeaker output when terminated in a resistive load equal to the normal loudspeaker impedance.

3.5.2.6 Full rated output. The full rated output of the receiver shall be at least two watts into a resistive load equal to the nominal loudspeaker impedance.

3.5.2.7 Audio distortion. The overall receiver audio distortion for a power output of two watts shall not exceed five percent.

3.5.2.8 Audio frequency response. The overall audio frequency response shall not vary more than -1 to +3 dB over the frequency range of 300 to 3000 Hz.

3.5.2.9 Hum and noise. The hum and noise level shall be at least 50 dB below the full power output level of the receiver.

3.5.2.10 Cross modulation. The simultaneous application of a signal 1.5 MHz off resonance at 0.1 volt and an in-channel signal at any level from 1.5 microvolts (μv) to one volt shall not produce more than 0.05 watts in the audio load.

3.5.2.11 Modulation acceptance bandwidth. The modulation acceptance bandwidth shall be greater than $\pm$5 kHz.

3.5.2.12 Selectivity. The bandwidth of the IF amplifier shall be within the following limits with respect to the center frequency:

Attenuation	Bandwidth
6 dB	$\pm$ 10 kHz minimum
40 dB	$\pm$ 15 kHz maximum
60 dB	$\pm$ 32 kHz maximum

3.5.2.12.1 Adjacent channel selectivity and desensitization. The adjacent channel selectivity shall, as a minimum, be 70 dB.

3.5.2.13 Undesired response. The ratio of any undesired input voltage to the input voltage at resonance producing one watt power output shall be at least 60 dB. Undesired is any frequency outside the 60 dB passband specified in 3.5.2.12.

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3.5.2.14 Image response. Response to image frequencies shall be at least 70 dB below that of the carrier frequency.

3.5.2.15 Spurious response. Response to spurious frequencies shall be at least 70 dB below that of the carrier frequency.

3.5.2.16 Useable sensitivity. The useable RF sensitivity shall be no greater than two microvolts (with 50-ohm input impedance) for 10 dB signal-plus-noise-to-noise ratio at one-half the full audio power output level across a resistive load equal to the loudspeaker impedance.

3.5.2.17 Quieting sensitivity. The quieting sensitivity shall be no greater than two microvolts for the same conditions specified in 3.5.2.16.

3.5.2.18 Frequency stability. The center of the passband shall not vary from the nominal tuned channel frequency more than ± 0.0005 percent under the environmental conditions specified in 3.4 through 3.4.4.

3.5.2.19 Intermodulation spurious attenuation. The intermodulation spurious attenuation shall be at least 50 dB.

3.5.2.20 Automatic gain control. The receiver shall have an automatic gain control which will maintain audio output that will not vary more than 4 dB when the RF input voltage in the channel frequency is varied from 10 μv to 100 millivolts (mv).

3.5.2.21 Squelch. A squelch circuit shall be provided. The threshold shall be adjustable and capable of being disabled for both AM and FM modes of operation.

3.5.3 Transmitter.

3.5.3.1 Transmitter frequency range. The transmitter shall be designed for AM and FM operation at those frequencies listed in 1.1. Twenty-five kHz channel selector increments are to be used in all ranges. All transmitter specification requirements shall be conformed to for every frequency in every range.

3.5.3.2 Rated power output. The unmodulated RF carrier power output shall not be less than five watts into a 50-ohm resistive load.

3.5.3.3 Power output stability under environmental conditions. At any of the specified operating line voltages, the power output shall not be below five watts under the environmental conditions specified in 3.4 through 3.4.4.

3.5.3.4 Carrier stability. At rated power output, the unmodulated carrier frequency shall be stable within ±0.0005 percent under the environmental conditions specified in 3.4 through 3.4.4.

3.5.3.5 Carrier distortion. For audio input frequencies from 300 to 3000 Hz, the carrier distortion shall be no greater than 10 percent.

3.5.3.6 Spurious output. At full power output the level of all spurious outputs as measured at the output connector shall not exceed a level of 60 dB below the level of the carrier fundamental.

3.5.3.7 Carrier noise. At full power output, the noise level on the carrier, unmodulated, measured at the output of the transmitter terminated in a 50-ohm resistive load shall be at least 60 dB below the modulated carrier level.

3.5.3.8 AM hum and noise. AM hum and noise shall be attentuated at least 40 dB.

3.5.3.9 Modulation (AM). A method shall be provided to prevent the modulation level from exceeding 100 percent.

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3.5.3.10 Modulation capability (AM). The carrier shall be capable of being modulated 95 percent within the audio range of 300 to 3000 Hz.

3.5.3.11 Modulation control (AM). The screwdriver-adjustable control shall be provided for the adjustment of modulation percentage. This control and associated circuitry shall provide sufficient gain to modulate the transmitter 95 percent with a quiet-speaking voice into the microphone and provide reduction of gain to prevent over-modulation with a louder-speaking voice.

3.5.3.12 FM characteristics.

3.5.3.12.1 Modulation. Frequency modulation shall be employed. The system shall be designed for adjustable frequency deviation to ±7.5 kHz.

3.5.3.12.2 FM hum and noise level. FM hum and noise shall be attenuated at least 40 dB.

3.5.3.12.3 Modulation limiting. The instantaneous peak and steady-state deviations shall not exceed rated system deviation at any audio frequency or change in input level.

3.5.3.13 Harmonic distortion. The total harmonic distortion shall not exceed five percent at full rated power output.

3.5.3.14 Audio frequency response. The audio frequency response shall not vary more than +1 or -3 dB over the frequency range of 300 to 3000 Hz.

3.5.3.14.1 Response below 300 Hz. Response from zero to 300 Hz shall be attenuated not less than 12 dB per octave.

3.5.3.14.2 Response above 3000 Hz. Response shall be attenuated not less than 12 dB per octave for frequencies from 3000 to 20,000 Hz, and not less than 6 dB per octave from 20,000 to 30,000 Hz.

3.5.4 Frequency control.

3.5.4.1 Radio frequency synthesizer. A frequency synthesizer shall be provided as part of the transceiver. All channel frequencies shall be derived from the use of a single crystal. The synthesizer shall be common to both the receiver and the transmitter of the radio set.

3.5.4.1.1 Synthesizer temperature control. The design of the frequency synthesizer shall not require the use of a temperature controlled compartment.

3.5.4.2 Receiver-transmitter tuning. The receiver and transmitter shall be easily tuned to any 25-kHz channel in the specified frequency ranges (see 1.1). All requirements shall be conformed to for every channel frequency without the need to readjust the radio set.

3.5.4.2.1 Alignment tool kit. Any special tools required for service alignment of the transceiver shall be provided with each radio set. Special tools are those not listed in the Federal Stock Catalog.

3.5.4.2.2 Preset frequencies. The synthesizer shall be designed so the transceiver is capable of presetting or preprogramming at least four frequencies so that these fequencies may be accessed immediately without having to reset the main frequency control dial(s).

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3.5.5 Power supply. The transceiver shall operate on each of the following: 115 volts AC at 60 Hz, 12 VDC and 24 VDC external to the unit, and on the rechargeable nickel-cadmium battery contained within the unit. A built-in charger shall be provided so that the battery can be charged from 115 VAC, a 12 VDC vehicular battery, or a 24 VDC vehicular battery with the engine running or not running. The battery shall be on charge during all operation from an external source. The battery charged from a vehicle shall provide the same capabilities as when charged from an AC source. Built-in protection shall be provided to prevent damage to the battery due to overcharge. The charge rate shall be such as to fully charge a completely discharged battery within 16 hours; the charging rate shall be such that on continuous charge no battery or transceiver deterioration or failure shall result regardless of battery condition.

3.5.5.1 Battery. A front panel indicator shall be provided to indicate the charge condition of the battery during portable operation. The nickel-cadmium battery shall have sufficient capacity for a minimum of eight hours operation throughout the duty cycle of 10 percent receive, 10 percent transmit with 1000 Hz, 0.5 watts audio output, and 80 percent standby. When operated from an external power source, the self-contained battery shall be capable of being disconnected and removed from the unit without affecting operation of the transceiver. Disconnection and removal of the battery shall be simple and without the use of tools.

3.5.5.2 Power switch. One ON/OFF power switch shall be provided on the front panel for controlling the AC or DC power input as appropriate to the equipment. The power switch shall also provide a battery charge position, whereby the battery can be charged while the transceiver is turned OFF.

3.5.5.3 Power connector. A single connector shall be provided for connecting either the AC or DC power cable specified in 3.3.6 to the transceiver. The connector shall have contacts which shall provide the required selective connection for the AC or DC input. The grounding wire of the AC power cable shall ground the metal chassis and framework of the transceiver.

3.6 Reliability. The equipment shall have an upper limit mean-time-between-failure (MTBF) test (θ_0 as defined in MIL-STD-781) of 1500 hours and a lower limit MTBF test (θ_1 as defined in MIL-STD-781) of 750 hours.

3.6.1 Equipment conditioning. The equipment shall be capable of withstanding the equipment conditioning test of 4.5.4.

3.7 Maintainability. The equipment shall be capable of demonstrating the limits specified in 3.7.1 and 3.7.2 for corrective maintenance in compliance with MIL-STD-471.

3.7.1 Quantitative corrective maintenance. The equipment, including built-in-test (BIT) and Government furnished equipment/property (GFE/GFP), shall have a mean-corrective-maintenance time, (M_{ct}) not exceeding 0.5 hour and a maximum-corrective-maintenance time (M_{maxct}) (95th percentile) not exceeding 1.0 hour when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, printed circuit (PC) boards) and chassis-mounted parts (electronic, electrical, electro-mechanical, and mechanical parts). The corrective-maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective-maintenance actions. BIT circuits are included in the corrective-maintenance time requirement. The equipment is considered to be in a corrective-maintenance mode whenever a failure or a malfunction occurs, including a malfunction that allows degraded operation.

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3.7.2 Subassembly (module) maintainability. Repairable modules shall have a M_{ct} of 2.0 hours and a M_{maxct} of 4.0 hours (95 percentile) at the organizational, intermediate, or depot-level maintenance as appropriate. M_{ct} for module repair includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all maintenance tasks. For purposes of maintainability, a module is considered the lowest replaceable subassembly (PC card) for which repair is accomplished by the replacement of piece parts.

3.8 Safety. Safety shall be in accordance with MIL-STD-454, as specified in MIL-E-16400.

3.8.1 Leakage current. Equipment leakage current shall not exceed five milliamperes (mA) AC or DC.

3.9 Thermal design. Thermal design shall be in accordance with Requirement 52 of MIL-STD-454.

3.10 Electromagnetic compatibility (EMC). The equipment shall be self-compatible and shall conform to the requirements of Classes IA and IB of MIL-STD-461, with the following exceptions:

a. CE01: Broadband need not be performed.
b. RS03: Shall be performed at 50 volts per meter from 14 kHz to 30 MHz and 10 volts per meter from 30 MHz to 10 GHz.
c. CS06: Shall be performed at 400 volts peak and 5 μs.
d. CE06: Key-up, no change.
e. CE06: Key-down limits shall be in accordance with 3.5.3.6 and 3.5.3.7.
f. CS03: Limits shall be in accordance with 3.5.2.19, with the two intermodulated signals injected at a level of 0.1 volt rms.
g. CS05: Limits shall be in accordance with 3.5.2.10.
h. CS07: Modes shall be in accordance with 3.5.2.21 and to the limits specified in MIL-STD-461.
i. CS08: Limits shall be in accordance with 3.5.2.12, 3.5.2.13, 3.5.2.14, and 3.5.2.15.
j. RE02: Limit at transmitter tuned frequency is exempt.

3.11 Workmanship. The workmanship of the equipment shall conform to the best commercial standards and shall be capable of withstanding a defect detection vibration screen of random type vibration at $0.01g^2/HZ$ $\pm$3 dB from 100 to 500 Hz.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

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4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection
   1. Production inspection (see 4.4.1)
   2. Production control inspection (see 4.4.2)
c. Environmental inspection (see 4.4.3)
d. Reliability testing (see 4.6)
e. Maintainability demonstration (see 4.7)
f. Inspection of preparation for delivery (see 4.8)

4.2.1 Test conditions. Unless otherwise specified, tests shall be performed at factory ambient temperature, humidity, and pressure. Input power shall be 115 volts ±10 percent at 60 Hz ±5 percent.

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE I.

TABLE I. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article	Quality conformance Group A	Group B	Group C
Surface examination	3.2.3	4.5.1	X		X	
Weight	3.2.4	4.5.1	X		X	
Size	3.3	4.5.1	X	X		
Parts and materials	3.3.4.3	4.5.1	X	X		
Finish	3.3.4.2	4.5.1	X	X		
Marking	3.8	4.5.1	X	X		
Safety	3.11	4.5.1	X			
Workmanship	3.11	4.5.2	X		X	X
Workmanship screen	3.5	4.5.3.1	X	X	1/	1/
Pre-performance	3.5	4.5.3	X	X		
Performance	3.6.1	4.5.4	X			X
Equipment conditioning	3.4.1	4.5.5.2	X			X
Low temperature	3.4.1	4.5.5.1	X			X
High temperature	3.4.2	4.5.6	X			
Humidity	3.3.1, 3.3.4.2	4.5.9	X			X
Salt fog	3.4.4	4.5.8	X			X
Shock	3.4.3	4.5.7	X			
Vibration	3.3.4	4.5.10	X		X	
Enclosure					X	
Steady-state voltage and frequency	3.5.1	4.5.11	X		X	
DC voltage variation	3.5.1	4.5.11	X			X
Power interruption	3.5.1	4.5.11	X		X	
Leakage current	3.8.1	4.5.12	X			
Electromagnetic interference	3.10	4.5.14	X		X	
Thermal design	3.9	4.5.13	X			
Reliability test and demonstration	3.6	4.6	X			
Maintainability demonstration	3.7	4.7	X			

1/ See TABLE II.

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4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process, such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests and examinations of TABLE I, Group A.

4.4.2 Production control inspection. Production control inspection shall be conducted on a sampling basis in accordance with the procedures of MIL-STD-105, using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall include those examinations and tests shown in Group B of TABLE I to an acceptance quality level (AQL) of 6.5 percent and shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily conform to the requirements of production control inspection prior to release for shipment.

4.4.3 Environmental inspection. Environmental inspection shall be as listed in TABLE I, Group C, and shall be performed on units which have been subjected to and passed production inspection.

4.4.3.1 Environmental inspection samples. Samples for the environmental inspection shall be selected without regard to their quality, in accordance with the following:

a. One sample from the first 25 production units fabricated
b. One sample from each successive 100 units, or fraction thereof, produced

4.5 Test methods.

4.5.1 Surface examination. The equipment shall be subjected to visual and functional inspections for:

a. Workmanship, assembly and fit, and mechanical safety for compliance with the requirements of 3.2
b. Materials, parts, finish, coatings, treatments, and marking for compliance with the requirements of 3.3

4.5.2 Workmanship screen. Prior to conducting the equipment conditioning test, vibration shall be performed on each equipment. The vibration shall be performed at the module or equipment level. All the hardware, including cables and connectors, shall be exposed to vibration. The vibration shall be random, or subject to procuring activity approval, pseudorandom or complex waveform vibration, for ten minutes in the axis deemed most susceptible to vibration excitation. All test items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 1. The control accelerometer shall be located next to one of the mounting points of the item under test. The test item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. Any failure shall require corrective action and test repetition.

4.5.3 Performance test. The equipment shall be energized and subjected to a performance test conducted in compliance with MIL-E-16400 to ensure the proper functioning and performance at an adequate number of points in the required range to demonstrate conformance to specification requirements, including safety. The performance test shall include, but not be limited to, the tests shown in TABLE II.

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NOTE: The acceleration spectral density (ASD) shall be controlled by an accelerometer signal with a filtered bandwidth of 10 Hz up to a vibration frequency of 500 Hz and a 100 Hz bandwidth above 500 Hz.

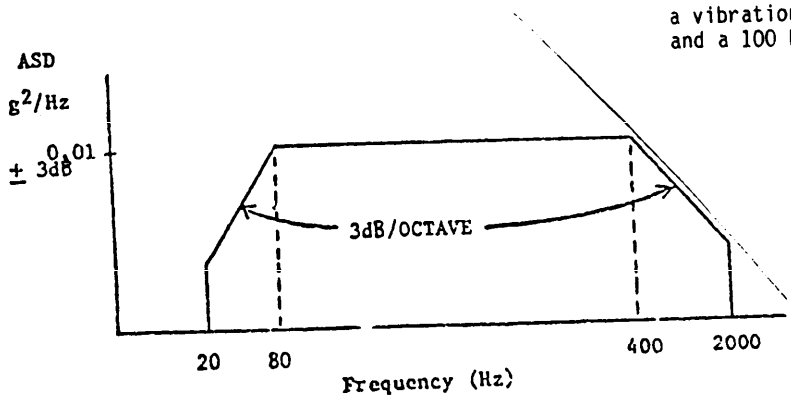


FIGURE 1. Random vibration curve.

TABLE II. Performance test.

Test	Requirement paragraph	Test paragraph	First article	Quality conformance Group A	Quality conformance Group B and C
Rcvr output	3.5.2.4 1/	4.5.4	X	X	
Rcvr sensitivity	3.5.2.16 and 3.5.2.17 1/	4.6.3	X	X	
Rcvr RF intermodulation	3.5.2.19	4.6.3	X	X	
Rcvr cross modulation	3.5.2.10	4.6.3	X	X	
Rcvr frequency stability	3.5.2.18 1/	4.6.3	X	X	
Rcvr additional characteristics	3.5.2.1 through 3.5.2.21 1/	4.5.4	X		X
Xmttr rated power output	3.5.3.2 1/	4.5.4	X	X	
Xmttr carrier stability	3.5.3.4 1/	4.6.3	X	X	
Xmttr spurious output	3.5.3.6	4.5.11	X	X	
Xmttr modulation capability	3.5.3.10, 3.5.3.12.1 through 3.5.3.12.3	4.6.4	X	X	
Xmttr additional characteristics	3.5.3.1 through 3.5.3.14	4.5.4	X		X
Rcvr-Xmttr tuning	3.5.3.2	4.6.4	X		X
Synthesizer preset	3.5.4.2.2 1/	4.6.3	X	X	
Power supply 115 VAC 60 Hz	3.5.5	4.5.11	X	X	
Power supply 12 and 24 VDC	3.5.5	4.5.11	X	X	
Power supply battery	3.5.5.1	4.5.11	X		X

1/ See 4.5.3.1.

4.5.3.1 Pre-performance test. The pre-performance test shall be conducted in accordance with MIL-E-16400 and shall include those requirements designated 1/ in TABLE II, as a minimum.

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4.5.4 Equipment conditioning. Each production unit shall be subjected to an equipment conditioning test as follows:

a. Duration 48 hours with the last 24 hours failure-free. If failure occurs during the last 24 hours, appropriate corrective action shall be implemented and the test continued until 24 continuous hours of failure-free operation have been accrued.
b. Monitor transmit and receive power output
c. Operate transmitter and receiver at full power into dummy loads
d. Cycle temperature as in the reliability test (see 4.6.1c)

4.5.5 Temperature tests.

4.5.5.1 High temperature. The equipment shall be subjected to test Method 501.1, Procedure I, of MIL-STD-810. The high-temperature nonoperating level shall be +71°C. The high temperature operating level shall be 60°C.

4.5.5.2 Low temperature. The equipment shall be subjected to tests specified in Method 502.1, Procedure I, of MIL-STD-810. The storage test shall be a minimum four hours soak at -57°C (-70°F). Operating tests shall be made at -30°C (-22°F).

4.5.6 Humidity. The equipment shall be subjected to test Method 507.1, Procedure II, of MIL-STD-810 with performance measurements made between the 43rd and 48th hour of each cycle.

4.5.7 Vibration. The equipment shall be subjected to resonance search, dwell, and cycling vibration tests in accordance with Method 514.2, Procedure VIII, of MIL-STD-810, using test curve W. The equipment shall be energized and fully operating during the test.

4.5.8 Shock. The equipment shall be subjected to Method 516.2, Procedure I, of MIL-STD-810, for ground equipment. The equipment shall be energized and fully operating during and after the shock test and shall be capable of withstanding a shock spectra of Figure 516.2-1 of MIL-STD-810 applying 10 shocks per axis at a peak value of 20g.

4.5.9 Salt fog. The equipment shall be subjected to Method 509.1, Procedure I, of MIL-STD-810 for sheltered equipment. The test shall be applied to the finishes and coatings on parts and frame and enclosure structures as finally assembled for use. Sample corner structures and any other critical sections may be used for the test. The test shall not be applied to the complete equipment.

4.5.10 Enclosure test. Enclosures shall be subjected to the dripproof test of MIL-STD-108. The actual 15-degree dripproof test shall be conducted on this equipment, and for first article inspection the enclosure test shall be performed after the completion of the vibration and shock tests.

4.5.11 Supply line voltage and frequency. The equipment shall be subjected to the following supply line tests of MIL-E-16400:

a. Steady-state voltage and frequency (115 VAC 60 Hz)
b. Voltage variation (12 and 24 VDC)
c. Power interruption

4.5.12 Leakage current.

WARNING: THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

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4.5.12.1 Equipment connections. Leakage current shall be measured on equipment in its normal operating configuration. Each equipment directly connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipment and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment under test and the source power shall be opened. If the equipment must be touched during the test, personal protective devices (for example, rubber gloves, insulating tools, or knobs), shall be used for protection against electric shock. The equipment shall be connected as shown in FIGURE 2 if it is connected to single-phase or DC source power.

4.5.12.2 Measurement. The leakage current shall be determined by the voltage-drop method. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 volts at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the equipment is designed. The overall measurement error shall not exceed five percent. A True RMS Voltmeter shall be used. The probe shall be connected to the equipment enclosure and associated unit enclosures and measurements taken for every combination of switch positions available in the test diagram. The probe shall also be connected to all external conducting parts such as connector housings, case, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in FIGURE 2. Equipments having more than one mode of operation, such as transmit and receive, shall be tested in each mode. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.13 Thermal test. The equipment shall be subjected to the thermal test of MIL-E-16400 to demonstrate compliance with the thermal design requirements of 3.9.

4.5.14 Electromagnetic interference. The equipment shall be subjected to the electromagnetic interference tests of MIL-E-16400 to demonstrate conformance to the requirements of 3.10.

4.6 Reliability test. The first article shall be tested in accordance with the requirements of Test Plan IVC of MIL-STD-781.

4.6.1 Environment. The environment for the reliability test shall be as follows:

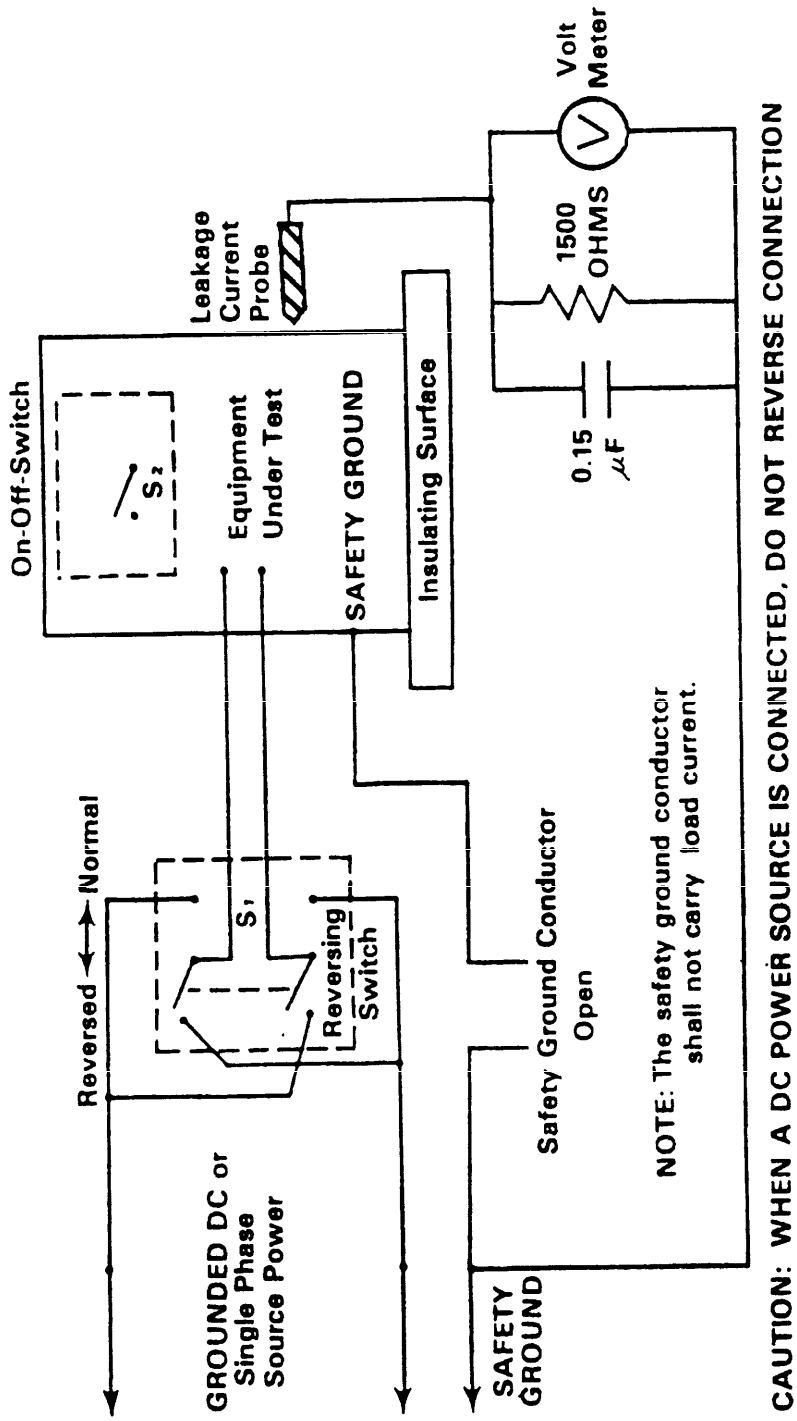
a. Electrical stress. Nominal voltage for 50 percent of the time and minimum and maximum voltage for 25 percent of the time, respectively.
b. Vibration. Swept sine vibration shall be applied over the frequency range of 5 to 500 Hz for 15 minutes per hour of operation, as follows:
   1. 25.4 millimeters (mm)(1.0 inch) double amplitude 5 to 6.3 Hz, 2g from 6.3 to 500 Hz.
c. Thermal stress. The temperature shall be cycled between -30°C and +60°C in accordance with FIGURE 3.
d. Moisture. Moisture levels sufficient to cause visible condensation and frosting shall be used. Humidity levels need not be held constant. Moisture may be injected during appropriate times during the test cycle.

4.6.2 Failure. Failure occurs any time the equipment fails to conform to any of the requirements specified in Section 3.

4.6.3 Performance monitoring. Performance monitoring shall be as follows:

a. Performance measurements shall be made once during each eight-hour cycle and shall be made at the temperature extreme then current in the cycle. Approximately one- half of the performance monitoring shall occur at high temperature and one-half at low temperature. Frequency shall be changed daily in one-octave intervals until the whole frequency band is transversed; the cycle shall then be repeated until the test terminates.
b. The following parameters shall be monitored every eight hours during the test:
   1. RF power output. Unmodulated
   2. RF carrier output. Terminated in a 50-ohm resistive load
   3. Carrier stability
   4. Modulation capability
   5. Receiver output
   6. Receiver sensitivity
   7. Frequency stability

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WARNING: THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

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FIGURE 2. Single phase or DC test diagram for leakage current measurements.

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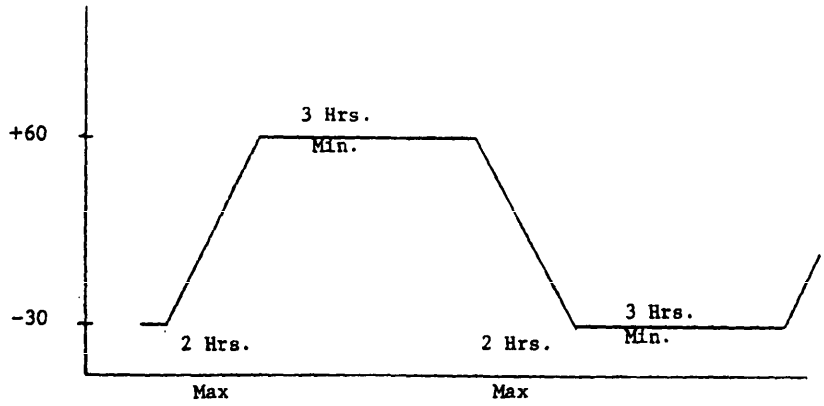


FIGURE 3. Thermal stress.

4.6.4 Operation during test. At all times other than when performance parameters are being monitored, the transceiver shall be operated as follows:

a. Receiver output terminated in a loud speaker or equivalent load with the gain control set at the half power point
b. Transmitter output terminated in an antenna or equivalent complex impedance load
c. The duty cycle shall be nine minutes receive and one minute transmit. During the one minute of transmit the transmitter shall be modulated 100 percent with a 1000-Hz continuous wave (CW) signal.
d. The frequency shall be changed daily in one-octave intervals until the whole frequency band has been transversed. The cycle shall be repeated until the test is terminated.

4.7 Maintainability demonstration.

4.7.1 Maintainability equipment demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be performed by qualified technicians and shall be used to verify that the equipment corrective maintenance requirements are fulfilled (see 3.7).

4.7.1.1 Demonstration conditions. The equipment maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual PC boards) and chassis-mounted electronic, electrical, electro-mechanical, and mechanical components or parts at the organizational level. One hundred candidate corrective maintenance tasks shall be identified (selected in accordance with Appendix A of MIL-STD-471) and assigned numbers of 1 to 100. The following information shall be made available for each candidate task:

a. Designation of specific faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part or simulation thereof)

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Faults of BIT functions shall be included as corrective maintenance tasks for the maintainability demonstration. The procuring activity or its authorized representative shall use the candidate tasks as a guide to select a sample of 50 tasks for the demonstration.

4.7.2 Accept/reject criteria.

4.7.2.1 Corrective maintenance. The accept/reject criteria for the demonstration of the corrective maintenance times for the demonstration shall be as indicated in TABLE III:

TABLE III. Acceptance criteria

	M_{ct}	M_{maxct}
Acceptance level	20	0
Sample size	50	50

Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value of each specified index (M_{ct}, M_{maxct}) is less than or equal to that shown in the acceptance table corresponding to each index. The duration of each task shall be compared to the required value(s) and recorded as greater than or lesser than each index. The actual value shall also be recorded. An accept decision can only be made for both M_{ct} and M_{maxct}. Accept/reject criteria for preventive maintenance shall be the capability of being able to perform each preventive maintenance task with no degradation of system performance. If degradation of system operations exists, the preventive maintenance test will have resulted in a reject decision.

4.7.3 Preventive maintenance. The demonstration shall qualitatively demonstrate that on-line preventive maintenance for the equipment shall not interfere with on-line operation.

4.7.4 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the equipment will be installed.

4.7.5 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the maintainability demonstration.

4.7.6 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation delivered with the equipment.

4.7.7 Rejection. Failure to conform to any of the requirements of this specification shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies
b. Upon approval of the approach, the contractor shall repeat the demonstration until an accept decision is reached.

4.7.8 Acceptance. The maintainability demonstration shall exhibit that the equipment conforms to the accept criteria prior to any delivery.

4.8 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance with the requirements of Section 5.

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The equipment described in this specification is intended for shore use only in non-tactical communications between field operator and air control facility and aircraft, as well as airfield emergency communications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3.
c. Levels of preservation, packaging, packing, and marking required (see 5.1).

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104. 55 of the Defense Acquisition Regulation (DAR). The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

Preparing activity:
NAVY-EC
(Project 5820-N148(EC))

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